

Adaptive Frequency Hopping (AFH) Mechanism in BLE

ECSE 4560/6560 Modern Communication Systems

Project Midterm Report

11/07/2025

Anthony Frias, friasa@rpi.edu

Hayden Fuller, fulleh@rpi.edu

1. Abstract

This project aims to model and simulate the Adaptive Frequency Hopping (AFH) mechanism in Bluetooth Low Energy (BLE) using MATLAB. The primary goal is to analyze how AFH improves communication robustness in interference-prone environments, particularly within the crowded 2.4 GHz ISM band, which is shared by various wireless technologies, namely Wi-Fi and Zigbee.

BLE is widely used in low-power applications such as wearables, medical devices, and IoT nodes. However, its operation in the 2.4 GHz band exposes it to heavy interference. AFH enhances BLE's coexistence capabilities by detecting busy channels and dynamically avoiding them. By simulating AFH, we aim to understand its effectiveness in maintaining performance and reducing packet loss in the presence of interference, thus contributing to the design of more reliable wireless communication systems.

2. Evaluation Plan

The project will simulate the BLE frequency hopping mechanism and introduce a simplified AFH algorithm. The model will include:

- The BLE channel map (40 RF channels)
- Interference modeling (Wi-Fi, BLE, and other noise sources)
- Channel quality assessment based on Packet Error Rate (PER) or simulated noise levels
- Dynamic channel blacklisting and whitelisting (AFH update)
- Comparison of system performance with and without AFH

Performance metrics such as PER, throughput, channel utilization, and latency will be evaluated under varying interference conditions. All simulations and analysis will be carried out using MATLAB because of its signal processing capabilities. Our simulation will be built with these key components:

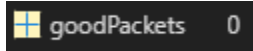
- BLE Base Simulation Setup:
 - Simulate BLE frequency hopping across 40 channels (2 MHz spacing).
 - Model periodic packet transmission and basic GFSK modulation.
 - Include an interference generator that can randomly or selectively impair certain channels.
- AFH Mechanism Implementation:
 - Monitor channel performance over time (using simulated PER or SINR).
 - Dynamically update a channel map that flags poor-quality channels as "unavailable".
 - Perform frequency hopping only over available (whitelisted) channels.
- Performance Metrics:
 - Packet Error Rate (PER): Track lost or corrupted packets.
 - Throughput: Measure effective data delivery rate over time.
 - Channel Utilization: Analyze how many channels are in use before and after AFH adapts.
 - Latency: Estimate delay impact from channel re-selection or retransmissions.
- MATLAB Tools Used:
 - Custom scripts for simulation logic
 - MATLAB's built-in plotting tools for visualizing results (e.g., channel quality over time, PER trends)
 - Signal Processing Toolbox
- Testing & Validation Strategy:
 - Simulate multiple scenarios with different interference patterns (random, fixed-channel jamming, etc.)
 - Run baseline tests (static channel map) vs. AFH-enabled simulation

3. Progress so far & Preliminary Results

Implemented Components

- GFSK modulation/demodulation
- Channel map implementation (40 channels)
- Frequency hopping logic
- PER-based AFH blacklisting logic
- Channel timeout recovery

Initial Results:



We're still in the process of debugging our code and getting it into a fully functional state. While many features are implemented already, there are still underlying issues preventing us from finding any usable data.

This is a current work in progress, small bugs are acting as a massive roadblock, but once clear,, our code should quickly be ready to collect a large variety of useful simulation data.

Next Steps:

- Debug code to a fully functional state
- Introduce realistic interference (connections , disconnections and varying throughput on common Wi-Fi bands, other bluetooth devices, Zigbee, microwaves, etc)
- Expand simulation loops to analyze trends across thousands of packets.
- Automate plotting for PER and throughput vs. any number of factors, including interference that is time variable, evenly spread or clustered, growing or shrinking, etc.

4. Updated Timeline

Actionable Item	Timeline
Literature review on BLE and AFH Setup MATLAB environment	Accomplished
Implement basic BLE with simple frequency hopping over 40 channels Add interference modeling	Now-Nov 17 Work in progress, has been a roadblock debugging code
Perform baseline PER and throughput analysis without AFH Implement AFH Perform PER and throughput tests under a variety of conditions Make initial comparison	Nov 18-Nov20 Once the code is working properly this shouldn't take much time
Run extensive simulations to gather more data for statistical validation Analyze data Create visuals Summarize findings	Nov 20 - Dec 1

Finalize visuals and plots Build report Build presentation	Dec 1 - Dec 5
Final Presentation	Dec 5 - 12